

Chap 13. mini project

https://github.com/kkh3363/C_Lang

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01. Phone book

Mini project

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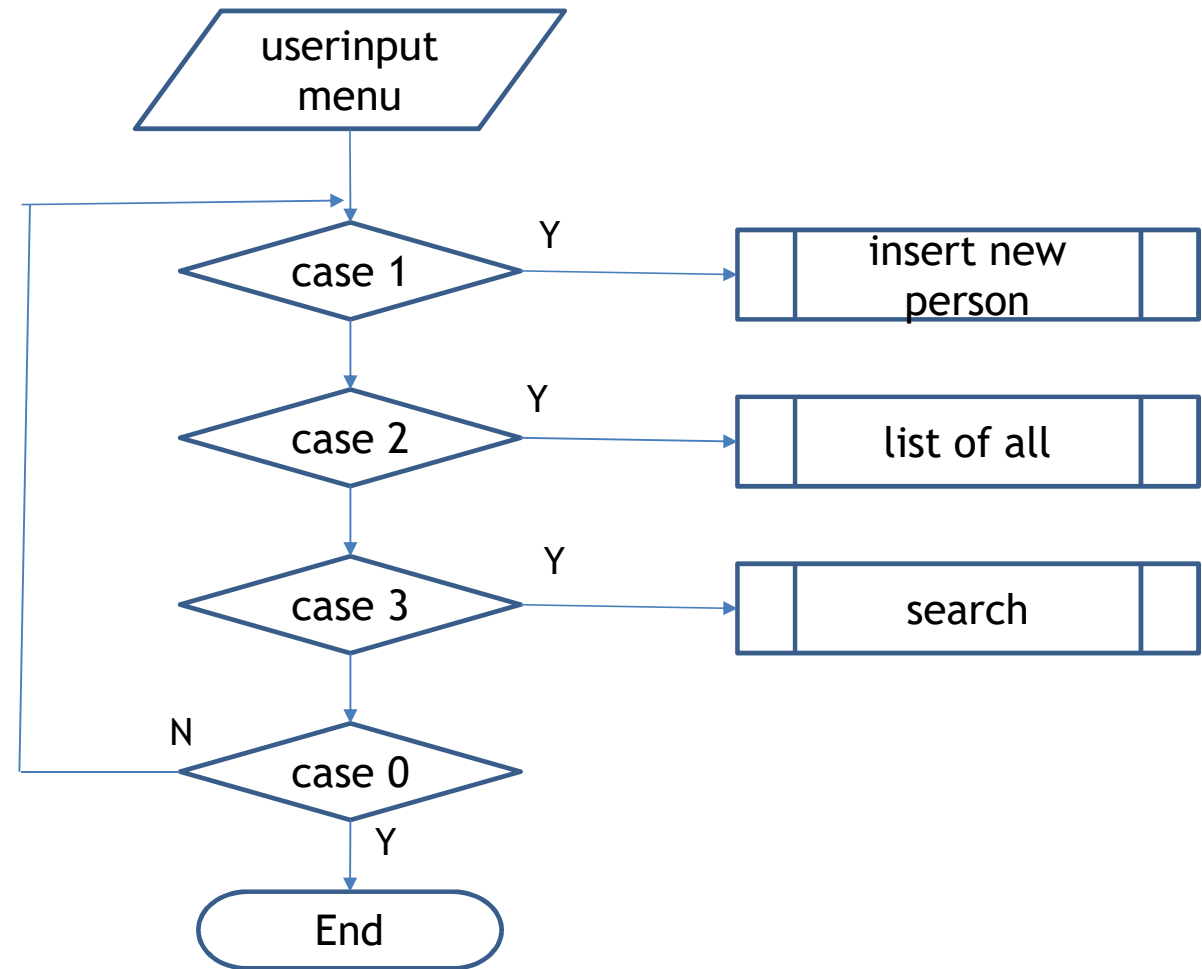
■ Intro.

- Let's programming a phone number management.
- program function is:
 - Insert user name, phone number
 - list all saved member's name, phone number
 - search phone number by name.

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■ Flow Chart



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- Create a function to receive menu input from the user

```
1 #include <stdio.h>
2 #include <string.h>
3 #define MAX 100
4
5 typedef struct {
6     char name[50];
7     char phone[20];
8 } Contact;
9
10
11 Contact contacts[MAX];
12 // index to input new user.
13 int count = 0;
```

```
15 int getMenu(){
16     int menu;
17     printf("\n===== My Phone book =====\n");
18     printf("1. Insert new name\n");
19     printf("2. list of all\n");
20     printf("3. Search phone number by name\n");
21     printf("0. Exit\n");
22     printf("Select menu number: ");
23     scanf("%d", &menu);
24
25     return menu;
26 }
27
```

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- insert new user

```
28 // insert new user
29 void addContact() {
30     if (count >= MAX) {
31         printf("Error: over storage.\n");
32         return;
33     }
34
35     printf("Name : ");
36     scanf("%s", contacts[count].name);
37
38     printf("Phone Number: ");
39     scanf("%s", contacts[count].phone);
40
41     count++;
42     printf("Success New user.\n");
43 }
```

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■ list of phone book

```
45 // list of phone book
46 void printContacts() {
47     int i;
48     if (count == 0) {
49         printf("List is Empty.\n");
50         return;
51     }
52     printf("\n=== List of Phone book ===\n");
53     for( i=0; i < count; i++) {
54         printf("%d. Name: %s / Phone-#: %s\n", i + 1, contacts[i].name, contacts[i].phone);
55     }
56 }
57
```


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- search phone num by name

```
58 // search phone num by name
59 void searchContact() {
60     char searchName[50];
61     int found = 0;
62     int i;
63     printf("Enter the name to search for: ");
64     scanf("%s", searchName);
65
66     for ( i = 0; i < count ; i++) {
67         if (strcmp(contacts[i].name, searchName) == 0) {
68             printf("=====\n");
69             printf("Found result:\n");
70             printf("name: %s\n", contacts[i].name);
71             printf("phone-#: %s\n", contacts[i].phone);
72             found = 1;
73             break;
74         }
75     }
76
77     if (!found) {
78         printf("I'm sorry. \n");
79     }
80 }
```

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■ main

```
81 int main(){
82     int menu;
83
84     do {
85         menu = getMenu();
86         switch (menu) {
87             case 1:  addContact(); break;
88             case 2:  printContacts(); break;
89             case 3:  searchContact(); break;
90             case 0:
91                 printf("Have a nice Day\n");
92                 return 0;
93             default:
94                 printf("Ooops. incorrect menu\n");
95         }
96     } while(1);
97     return 0;
98
99 }
```

- Run it to check the result.

Thank You !